

Lender heterogeneity in the pricing of bundled credit

Vittoria Iannotta*

October 8, 2026 - Draft

ABSTRACT

I document new empirical evidence on banks' pricing of bundles. Across 99 French lenders the average gap is -0.05 percentage points, close to zero, but it hides a spread of 1.69 points: nineteen lenders charge a premium, fifteen give a discount, and the rest show no significant gap. Which products a lender sells accounts for 4.9 percent of the variance across lenders, and the rating, size, age and leverage of its borrowers for 0.6 percent. The price gap does not evolve significantly over the bank-firm relationship. Therefore, most of the bundling strategy remains unexplained, and may rather be attributed to lender-specific strategies. This result highlights the strategic choices made by individual banks, and it opens several questions on borrowers' myopia and price discrimination.

*HEC Paris.

1 Introduction

Bundling products is a common practice in banking. In France, regulation governing bundling is particularly strict for retail consumers, while banks retain considerable discretion in their practices toward firms, including the pricing and composition of bundles. Despite the prevalence of bundling in banking, empirical evidence on the subject remains limited. In particular, little is known about how banks price bundles and which bundling strategies they adopt when serving firms.

While specific empirical evidence remains limited, the prevailing view is that banks offer bundles at an overall discount to attract and lock-in customers. These switching costs can subsequently allow banks to lock in customers and charge a premium on future products (Basten and Juelsrud, 2026, 2025). This effect is likely to be stronger when information asymmetries between banks and firms are greater, as relationship lending gives incumbent banks an informational advantage over potential competitors (Bharath et al., 2007).

I challenge this view by showing new empirical evidence on bundling behaviour of banks. First, I find substantial heterogeneity in the pricing of bundles across banks. Some banks offer bundles at a discount, while others charge a premium. Second, I show that this heterogeneity is not explained by differences in the characteristics of the borrowers served by each bank, nor by the composition of the bundles offered. Instead, it appears to be driven by lender-specific factors that are not directly observable in the data. The most direct implication of this finding is that the pricing of bundles is not solely determined by market forces, but also by the strategic choices made by individual banks. This also opens several questions on borrowers' myopia, as in Gabaix and Laibson (2006), and price discrimination.

I start by using detailed loan-level data from France. I restrict the analysis to a period of relatively stable rates, during the ZLB and before the pandemic. I identify a bundle whenever a borrower takes out multiple products from the same bank in the same month, and I compare the rate on the same product sold individually or in a bundle. I find substantial heterogeneity across banks: some systematically offer a discount and others charge a premium. Ninety-nine lenders write enough contracts on both sides to carry a gap of their own, and they hold 98% of the cross-sold records. Their gaps run from a discount of 0.81 percentage points to a premium of 0.88 percentage points. One common gap is rejected: Cochran's Q is 610 on 98 degrees of freedom, and the standard deviation of the true lender gaps net of estimation error, $\hat{\tau}$, is 0.16 percentage points.

I ask whether borrower composition and product mix are driving the observed heterogeneity. Credit quality barely affects the price of a bundle. Weak-rated borrowers pay 0.310 percentage points more for a product inside a bundle than alone (s.e. 0.139) and good-rated ones 0.144 (s.e. 0.045), while excellent, medium and ungraded borrowers

pay within 0.06 of zero. The price gaps estimated on these five risk classes are not all zero ($p < 0.001$) but cannot be told apart from one another ($p = 0.12$), and the whole schedule spans 0.30 percentage points against the 1.69 points separating the lenders.

While credit quality has a limited impact, the composition of the bundle matters more. An overdraft and a revolving credit line carry a premium inside a bundle, global operating credit and equipment leasing a discount, and 0.62 percentage points separate the dearest product from the cheapest.

I also ask whether bundles are used to open relationships. The credit register says whether a contract comes from a bank that was not lending to the firm the quarter before. First-time borrowers pay 0.179 percentage points more for a standalone product than continuing ones do (s.e. 0.014), and the bundle gap does not move with the relationship: the interaction is 0.021 (s.e. 0.057). Banks charge for a relationship they do not yet have, and they charge it on the product rather than on the bundle.

I decompose the share of bundle gaps explained by credit quality and product composition. For each lender I build the gap it would post if it charged every category what the market charges and differed from other lenders only in its mix - one prediction over the fourteen products, one over rating, size, age and bank-debt quartile - and split the variance of the lender gaps into the part moving with each prediction and the rest. Product mix accounts for 4.9% of that variance, the four borrower margins together 0.6%, and 94.5% is left over. The two mixes overlap little: scored in sample, each alone accounts for 6.9% and 1.3%, against 7.9% jointly. Subtracting both predictions from the gaps leaves their standard deviation at 0.29 percentage points. Inside the product mix, overdrafts carry 65% of the co-movement with the gap and revolving credit 22%; global operating credit carries a further 26% through the opposite sign, the dear lenders selling less of it, and the rest take thirteen points back off. Inside the borrower mix only rating carries anything, at 1.4%. The 94.5% left over is not a lender effect I measure. It is what the two mixes do not reach, and it holds whatever composition I do not observe together with the sampling error in the ninety-nine estimated gaps.

The last comparison holds the borrower fixed across lenders rather than holding its characteristics fixed. Of 911,105 firm-months, 24,587 hold two or more lenders, and 1,791 of those hold both a cross-sold and a standalone record: 1,285 firms and 9,089 records. On that support the average gap disappears - $\beta = 0.014$ with a standard error of 0.059 - so a firm taking a second product from one of its banks and not from another pays the same either way. The lenders disagree there by more, not less: the twenty-one carrying at least a hundred identifying records have gaps with a standard deviation of 0.56, Q is 68 on 20 degrees of freedom, and $\hat{\tau} = 0.32$ percentage points against 0.16 in the full sample. Holding the borrower fixed removes the average and leaves the dispersion.

2 Literature

Three papers come closest to this one, and each stops one step short of it. [Basten and Juelsrud \(2026\)](#) measure the price a bank charges a client it already has. [Bharath et al. \(2007\)](#) measure how often a bank wins that client's next product, and say they cannot observe its price. [Gabaix and Laibson \(2006\)](#) give the pricing structure - a subsidised base good beside a dear add-on - and an equilibrium in which every firm adopts it. All three deliver one number for the market. This paper delivers ninety-nine, one per lender, and rejects the common one.

Cross-selling and the price of the second product. [Basten and Juelsrud \(2026\)](#) follow every household-bank relationship in Norway from 2004 to 2018. A household that holds deposits at a bank is about 20 percentage points more likely to borrow there over the following fourteen years than a household in the same municipality, age, salary and education group. It then pays 15 to 27 basis points more on that loan, not less. The premium is not the price of better information: future defaulters pay 5 basis points more ex ante, and that penalty is 2 basis points smaller for existing depositors, so the bank screens its own clients no better than anyone else's. What holds them is inaction. Switching for a new loan becomes 5 percentage points less likely with each year of tenure, as in [Klemperer \(1995\)](#) and [Farrell and Klemperer \(2007\)](#). The discount sits on the first product instead: parents pay 14 basis points less on deposits and the under-thirties a further 28, and both groups borrow more later. [Basten and Juelsrud \(2025\)](#) price that option. A bank that expects its depositors to convert passes less of a policy rate cut into its deposit rate, by 10 to 16 percentage points of the pass-through, and one standard deviation of cross-selling potential moves the pass-through as much as one standard deviation of local concentration. Both papers read the variation across banks as variation in clientele: banks differ in the clients they hold, and therefore in the option they hold. Section 4.5 measures that reading and finds it accounts for 0.6% of the variance across lenders.

What a relationship is worth to the lender. [Bharath et al. \(2007\)](#) ask what the bank gets. A bank that has lent to a firm before supplies its next loan with probability 0.42; a bank that has not, with probability 0.03. The hold is tightest on firms outsiders cannot read: small firms and unrated ones. The mechanism is the one in [Kanatans and Qi \(2003\)](#), in which the cost of learning about a borrower is paid once and spread over every product sold to it, so an incumbent supplies any one product more cheaply than an entrant. Their constraint is the price. A bank that does not make a loan posts no rate, so they impute one for each bank in the choice set and enter it as a control. The register used here reports the all-in rate on every facility, and the same product appears inside and outside a bundle at the same lender in the same month, which is what makes the price of cross-selling observable. The sign that cross-selling puts on a loan rate is

contested. [Drucker and Puri \(2005\)](#) find a concession when a bank lends to a firm and underwrites for it at once, and [Lepetit et al. \(2008\)](#) find that European banks expanding into fee income charge less on loans. Against them, [Basten and Juelsrud \(2026\)](#) find a premium, and the euro-area supervisory survey in [Basten and Juelsrud \(2025\)](#) has the banks that price with cross-selling in mind paying more on deposits and charging more on loans, to firms as well as to households. Each of these papers settles the sign for one market with one number. Section 4.1 finds both signs inside one market at the same time, and the average of the two is -0.05 percentage points.

Add-ons, loss leaders and myopia. [Gabaix and Laibson \(2006\)](#) explain how a dear product survives competition. If enough buyers overlook the price of an add-on, every firm subsidises the base good and marks up the add-on, and no firm gains by unshrouding: the buyer it educates prefers to keep the subsidy and avoid the add-on. Financial services are their leading example, and the structure applies attribute by attribute, so one bundle can hold shrouded products beside salient ones. Section 4.3 reports that pattern: an overdraft and a revolving credit line cost more inside a bundle, global operating credit and equipment leasing less, and 0.62 percentage points separate the dearest product from the cheapest. The two dear products are the contingent ones, drawn after signing, whose rate a borrower need not confront when it chooses the bundle. Their equilibrium, though, is symmetric: every firm shrouds, or none does. It predicts the sign across products and nothing about the spread across lenders, which is the fact this paper reports.

Bundling in industrial organisation. [Adams and Yellen \(1976\)](#) show that a seller facing buyers whose valuations are negatively correlated earns more from a bundle than from its parts, and [McAfee et al. \(1989\)](#) give the conditions. [Whinston \(1990\)](#) and [Nalebuff \(2004\)](#) make the bundle a barrier to entry, and [Armstrong and Vickers \(2010\)](#) price bundles under competition. These models fix technology and demand and derive a price, so the bundle carries one sign in a market: a discount when it screens or deters, a premium when it exploits a captive second purchase. That common sign is the natural null for Section 4.1, and Cochran’s Q rejects it at 610 on 98 degrees of freedom.

Lender-level pricing. The dispersion documented here is a fact about lenders, not about products or borrowers, which puts it alongside work treating a price difference as something a lender chooses rather than something a market determines. [Degryse and Ongena \(2005\)](#) trace loan rates to distance and local competition, [Ioannidou and Ongena \(2010\)](#) to what a borrower pays to switch, and [Drechsler et al. \(2017, 2021\)](#) to deposit market power. The closest counterpart runs the other way: [Ongena et al. \(2021\)](#) document that mortgage rates are set borrower by borrower rather than at the level of the bank, so the price variation they study sits inside lenders rather than between them. None of them, and none of the four papers above, reports a cross-section of lenders pricing the same contract differently once the borrower is held fixed. Section 4.6 does: among

firms taking a second product from one of their banks and not from another, the average gap is 0.014 percentage points with a standard error of 0.059, and the twenty-one lenders identified there disagree by more than the full sample does.

3 Loan level data

The main dataset is a quarterly survey of all new loan contracts and credit lines to businesses in the first month of each quarter from 2006Q1 to 2024Q2 (“Nouveaux Crédits”, provided by Banque de France). It includes the all-in rate (TEG), the amount, the contractual maturity, an instrument code identifying the product, and the lending institution at the bank unit level; firm identifiers bring the Banque de France credit rating, total assets, employees, group affiliation, region and sector. The main limitation is that two months in three are never observed. I define a bundle as a firm taking two or more distinct products from the same bank in the same reporting month. The credit rating is assigned annually by Banque de France to firms and it represents a forward-looking assessment of creditworthiness over a three-year period. Firms are reviewed when their turnover exceeds €750K (€1250K from 2024). It can be observed by any lender before issuance. The credit rating scale has 12 rating categories, including one entry for liquidation (P) and one for when there is not enough information to assign a rating (0). A firm is evaluated on the basis of: the latest balance-sheet data available; data on default/delay events in payments (CIPE); sector-specific information; qualitative information, including interviews with firm managers on an annual basis, to assess the risks and difficulties the firm is facing.

Cleaning and sample. I restrict the window to 2016–2019, to focus on cross-sectional variations during the zero-lower-bound period, excluding Covid and the state guarantees (*prêt garanti par l’État*). I drop regulated and subsidised products, because on those the rate is not completely the lender’s choice. I aggregate multiple contracts into a single record if they share the same product, maturity, and rate within a firm-bank-month. The step merges 31% of contracts: 99.9% of other trade receivables, 81% of discounting, 73% of instalment sales finance, 65% of other treasury credit and 53% of Dailly assignments, against 1% of overdrafts. Table 1 gives the construction. The estimation sample is 1,087,737 records at 208 lenders, of which 72,379 are cross-sold. The control group is every other record: 848,007 that are the only facility in their firm-bank-month, and 167,351 that sit beside another record of the same product at a different price. Both are standalone borrowing in the sense that matters here, since the firm did not take a second product, so both belong on the comparison side. Graded borrowers are 340,213 of the records and ungraded ones 747,524. The dependent variable throughout is the TEG in percentage points, measured at the record level.

Table 1: From the register to the estimation sample

	Contracts
Contracts in the 2016-2019 window	2,529,651
after keeping non-regulated products	1,576,148
after aggregating identical facilities	1,087,737
of which cross-selling	72,379
of which control	1,015,358

Notes. Each row applies the restriction named to the row above. Regulated and subsidised products are dropped on the instrument code, and a contract is kept only if the *prêt réglementé* field reads “non”. Aggregation merges contracts sharing a product, a maturity and a rate within a firm-bank-month.

4 Empirical strategy

Cross-selling does not always come with a discount, and lenders disagree about what it costs. I measure the disagreement, then ask how much of it the products a lender sells and the borrowers it serves can account for. Neither accounts for much, and I say what the remainder is and is not.

4.1 Lenders price cross-selling differently

I start with a preliminary investigation. I compare a cross-sold contract with a standalone contract issued by the same lender, in the same month, in the same product category, at the same amount decile and maturity band, to a borrower with the same credit rating category and in the same sector. The sector enters as a sector-by-month effect inside the lender, so a contract is compared only with contracts the same lender writes that month to firms in the same industry. I estimate for each lender separately:

$$\text{TEG}_i = \beta_b \text{CrossSell}_i + \alpha_{c(i)} + \varepsilon_i, \quad i \in \mathcal{I}_b, \quad (1)$$

TEG is the total effective rate, including fees. CrossSell_i is one for a record in a firm-bank-month holding two or more distinct products and zero otherwise. I put a floor on the number of records per lender, so that each $\hat{\beta}_b$ is estimated with at least 500 records, 100 on each side, and a cross-sell share of at least 5% of its multi-record cells; 99 of the 208 lenders in the sample clear that floor, and they hold 98% of the cross-sold records. Standard errors cluster on the firm.

The coefficient of interest is the lender-specific gap $\hat{\beta}_b$. It compares the price of the same product within a bundle to the price of the product sold individually. The fixed effects make sure I am comparing products with similar terms across firms of comparable riskiness. Fig. 1 shows that lenders do not share a common bundle gap. The gaps run from -0.81 to $+0.88$ percentage points, with a weighted mean of -0.05 and a standard deviation across lenders of 0.29. Nineteen lenders charge a premium for cross-selling and

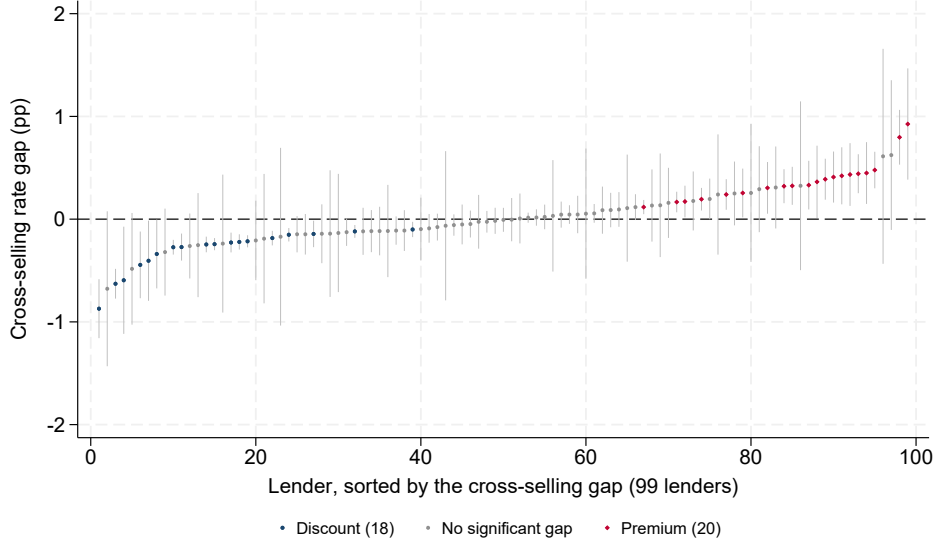


Figure 1: The cross-selling gap β_b of equation (1), lender by lender, sorted by the estimated gap, with 95% confidence intervals.

fifteen give a discount; the other sixty-five coefficients are not statistically different from zero.

I test whether the estimates are consistent with a single common gap measured with noise. Cochran’s Q does that, and $\hat{\tau}$ is the standard deviation of the underlying gaps net of estimation error. The test rejects a single common gap: $Q = 610$ on 98 degrees of freedom, $p < 0.001$, $I^2 = 84\%$ and $\hat{\tau} = 0.16$ percentage points.

Q reads the 99 estimates as independent draws, and they are not: 5.4% of the firms borrow at two or more of the 99 lenders and hold 15.2% of the records between them, so one firm’s rate shock enters several lenders’ gaps at once. Testing the same null in one regression that interacts the indicator with the lender, absorbs the same cell and clusters on the firm returns the same 99 gaps and rejects equality: $F(98, 361,664) = 6.6$, $p < 0.001$, on 706,996 records and 361,665 firm clusters. The dependence Q cannot see is not what produces the rejection.

Dropping the floor and estimating every lender the sample supports adds 18 more and leaves the conclusion intact ($Q = 631$ on 116 degrees of freedom, $\hat{\tau} = 0.16$), as shown in Appendix A. Putting every other column the register carries on the right-hand side takes $\hat{\tau}$ to 0.090 and still rejects a common gap, $Q = 292$ on 91 degrees of freedom (Appendix C).

I repeat the test for the twenty largest lenders by volume and the results are consistent: a standard deviation of 0.26, $Q = 84$ on 19 degrees of freedom, $p < 0.001$, and $\hat{\tau} = 0.15$. They are shown on their own in Appendix A, which also records the one place the test does not reject: holding the borrower fixed, the twenty largest lenders are consistent with a single common gap ($\hat{\tau} = 0.008$, $p = 0.43$), so the within-borrower disagreement comes

from the lenders outside them.

As far as the data allow, I add a firm fixed effect inside each lender’s own regression; this is effectively estimated on firms borrowing multiple times at the same bank. The median gap is estimated on 778 firms. The standard deviation of the coefficients $\hat{\tau}$ decreases to 0.083, and the test still rejects a common gap ($p < 0.001$).

There are different possibilities to explain this gap. The first is risk, coming either from borrower composition or from product mix. In the first case, the bank may simply recognise a riskier borrower and set a premium. In the second, the product mix matters, namely what each product is bundled with.

4.2 Borrower composition

One possibility is that the gap is driven by credit quality. As an example, if riskier firms are on average overpriced with respect to safer ones, the lender heterogeneity may be driven by borrowers’ sorting. The β_b estimated above is an average of class-specific gaps over lender b ’s borrower pool.

I exploit the rating categories: I group ratings into five classes, four from safest to riskiest and one for unrated firms. Then I estimate one gap per credit group on all lenders at once, absorbing the lender together with the interacted fixed effects above,

$$\text{TEG}_i = \sum_r \gamma_r (\text{CrossSell}_i \times \mathbf{1}[r(i) = r]) + \alpha_{\ell(i),c(i)} + \varepsilon_i, \quad (2)$$

so γ_r is the price of taking a second product as a borrower of class r , common across lenders. The variation behind each γ_r is one lender selling one product in one month to two borrowers of the same grade and the same sector, at the same amount decile and maturity band, one of whom takes a second product and one of whom does not. Standard errors are clustered at the firm level.

Table 2: The cross-selling gap by credit class, γ_r of equation (2).
Dependent variable: TEG, the all-in rate, in percentage points

	γ_r	s.e.	Cross-sold	Alone	Rate alone
Excellent	0.013	0.092	1,712	30,635	1.93
Good	0.144***	0.045	7,357	67,004	2.83
Medium	0.049	0.046	9,288	74,249	4.29
Weak	0.310**	0.139	1,936	27,343	7.92
Not rated	0.057***	0.014	50,388	548,018	7.40

Cross-sold and *Alone* count records on each side; *Rate alone* is the mean TEG of the standalone records. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

γ_r compares a cross-sold contract with a standalone contract to a borrower with the same rating and in the same sector, at the same lender, month, product, amount decile and maturity band. The cell already holds the rating, so γ_r does not capture the price

of risk. Credit quality barely moves the price of a bundle. Weak-rated borrowers pay 0.310 percentage points more for a product inside a bundle than alone (s.e. 0.139) and good-rated ones 0.144 (s.e. 0.045); excellent-rated borrowers pay 0.013 (s.e. 0.092) and medium-rated ones 0.049 (s.e. 0.046). The unrated, who hold seven of every ten cross-sold records, pay 0.057 (s.e. 0.014). The five gaps are not all zero ($p < 0.001$), but they cannot be told apart from one another ($p = 0.12$), and in proportion to what each class pays standalone the two that do clear zero are close: 0.310 on a weak borrower’s 7.9% is 3.9%, and 0.144 on a good borrower’s 2.8% is 5.1%. There is no gradient in risk here, and the whole schedule spans 0.30 percentage points - under a fifth of the 1.69 points separating the lenders in Figure 1. Nor does the gap fall on one kind of borrower rather than another inside a lender: Appendix B interacts the treatment with the borrower’s rating, age and bank dependence, and the only group paying a gap of its own is the one the register holds no creation date for.

4.3 Product composition

I repeat the same exercise with the product mix. As before, β_b is an average of gaps specific to bundle mixes that lender b offers. Different lenders may offer different product combinations, and this may explain the heterogeneity in the gap. I estimate one gap per product on all 99 lenders at once, absorbing the lender together with the cell,

$$\text{TEG}_i = \sum_p \gamma_p (\text{CrossSell}_i \times \mathbf{1}[p(i) = p]) + \alpha_{\ell(i), c(i)} + \varepsilon_i, \quad (3)$$

so γ_p is the price of taking product p inside a bundle rather than alone, common across lenders. γ_p compares a cross-sold contract with a standalone contract in the same product, at the same lender, month, amount decile, maturity band, rating and sector. Standard errors are clustered at the firm level.

Which product is in the bundle matters for the price of the overall bundle. For example, overdrafts and revolving credit lines come at a significant premium, while global operating credit and equipment leasing come at a significant discount. The dearest product and the cheapest are 0.62 percentage points apart.

Table 3: The cross-selling gap product by product, γ_p of equation (3).
Dependent variable: TEG, the all-in rate, in percentage points

	γ_p	s.e.	Cross-sold	Alone	Rate alone
Overdraft	0.224 ^{***}	0.042	10,480	321,854	11.11
Revolving credit	0.221 ^{***}	0.047	4,816	43,473	7.69
Personal loans	0.113 [*]	0.061	1,152	11,491	3.82
Instalment sales finance	0.104	0.068	239	5,841	4.48
Foreign receivables	0.094	0.127	1,588	3,128	2.02
Other equipment loans	0.070 ^{***}	0.008	21,449	191,020	1.92
Unregulated housing	0.041	0.032	1,286	16,551	2.20
Dailly assignment	0.026	0.057	5,419	22,979	2.43
Discounting	-0.008	0.088	2,573	17,057	2.27
Inventory finance	-0.061	0.052	916	4,217	2.03
Other treasury credit	-0.063	0.043	6,509	35,211	2.08
Global operating credit	-0.199 ^{***}	0.014	12,837	39,817	2.73
Leasing (equipment)	-0.273 ^{***}	0.085	1,146	28,433	5.27
Other credit to customers	-0.399 [*]	0.211	110	1,473	1.98

Sorted by γ_p . *Cross-sold* and *Alone* count records on each side; *Rate alone* is the mean TEG of the standalone records, in percent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.4 First-time borrowers

The loan data say what a lender writes in a month, not whether the borrower was already a client. The credit register does: it reports the credit outstanding for every firm-bank pair at the end of each quarter. I call a contract a first-time contract when the register shows the firm borrowing somewhere else in the previous quarter but not from that bank, and in a second version not from that bank in either of the previous two quarters. A firm the register does not record at all is not a continuing borrower: it has no flag and leaves the regression, because otherwise “I do not see this bank lending” and “I do not see this firm” would be the same event. That condition is what makes the flag readable on 52% of the records; a cross-sold record is readable more often than a standalone one, 62% against 51%, because two products at one bank clear the register’s reporting floor of 25 thousand euros more often than one does. On the readable records I estimate, pooling the lenders,

$$\text{TEG}_i = \beta \text{CrossSell}_i + \delta \text{New}_i + \theta (\text{CrossSell}_i \times \text{New}_i) + \alpha_{\ell(i),c(i)} + \varepsilon_i, \quad (4)$$

with the cell of equation (1) and standard errors clustered on the firm. δ is what a first-time borrower pays for a standalone product, β what a continuing borrower pays for

taking that product inside a bundle, and θ the difference between the two bundle gaps.

Table 4: The bundle gap at first-time and at continuing borrowers, equation (4).
Dependent variable: TEG, the all-in rate, in percentage points

	No credit in $q - 1$		No credit in $q - 1$ or $q - 2$	
	Coef.	s.e.	Coef.	s.e.
Cross-sell	0.017	0.016	0.014	0.016
New relationship	0.179***	0.014	0.199***	0.014
Cross-sell $\times$ new relationship	0.021	0.057	0.022	0.060
Cross-sell at a new relationship	0.038	0.056	0.036	0.058
Cross-sell, flag omitted	0.014	0.016	0.012	0.016
Records	453,279		435,474	

Left column: no credit from that bank in the previous quarter. Right column: none in the previous two. *Cross-sell at a new relationship* is $\beta + \theta$; *Cross-sell, flag omitted* is equation (1) on the same records. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First-time borrowers pay more, not less. A standalone product costs a first-time borrower 0.179 percentage points more than the same product costs a continuing borrower at the same bank in the same month (s.e. 0.014), and 0.199 (s.e. 0.014) on the two-quarter window. The bundle gap does not move with the relationship: $\theta = 0.021$ (s.e. 0.057), so the gap is 0.038 (s.e. 0.056) at a first-time borrower against 0.017 (s.e. 0.016) at a continuing one. Dropping the flag from equation (4) on the same records gives 0.014 (s.e. 0.016), so the restriction to readable records is not what holds the gap down. The descriptive side says the same thing: 9.5% of cross-sold contracts start a relationship against 27.4% of standalone ones, so bundles go to firms the lender already lends to.

Banks charge for a relationship they do not yet have, and they charge it on the product rather than on the bundle. Nothing here says that bundles are priced to open relationships. What the flag cannot see is a bank already lending below the register’s reporting floor, which it reads as a new relationship, so it dates the relationship from the first quarter above 25 thousand euros at that bank and the error runs toward calling continuing borrowers new rather than the reverse.

4.5 Product, borrower, or lender

Sections 4.2 and 4.3 price a credit class and a product, common across lenders. Neither says whether the lenders who look dear are the ones selling the dear products or the ones lending to the dear borrowers. I answer that with a shift-share.

Take any split of the cross-sold records into categories carrying a common gap, and write w_{bg} for lender b ’s share of its cross-sold records in category g . The weighted sum $\sum_g w_{bg} \gamma_g$ is the gap lender b would post if it charged every category the gap the market charges it and differed from other lenders only in its mix. I build two: P_b over the fourteen

products of Table 3, and B_b over rating, size, age and bank-debt quartile,

$$P_b = \sum_p w_{bp} \gamma_p, \quad B_b = \sum_r v_{br} \delta_r^R + \sum_s v_{bs} \delta_s^S + \sum_a v_{ba} \delta_a^A + \sum_d v_{bd} \delta_d^D. \quad (5)$$

The weights are counts, so they carry no sampling error, and all five sets of slopes come from one fit on 706,996 records. Which level each set omits does not matter: a CrossSell main effect carries the normalisation, so moving a reference shifts P_b and B_b by offsetting constants and leaves every covariance unchanged. Every lender helps estimate the slopes it is then scored against, so I refit the five sets 99 times, once dropping each lender, and quote the leave-one-out version throughout.

The variance of the gaps then splits in three,

$$\text{var}(\beta_b) = \text{cov}(\beta_b, P_b) + \text{cov}(\beta_b, B_b) + \text{cov}(\beta_b, \beta_b - P_b - B_b). \quad (6)$$

This is an identity for any two predictions, by linearity of covariance, so the three shares add to one exactly and no component is introduced before another. An incremental R^2 would not have that property: with the two mixes correlated, product-then-borrower and borrower-then-product give different answers, and neither is the decomposition. The third term is a residual, not a measured lender effect. It holds everything the two mixes miss, which includes unobserved composition and the sampling error in the 99 estimated gaps.

Table 5: Product, borrower, or lender: the three shares of equation (6), in percent of the variance of the lender gaps.

Across the 99 lenders of Figure 1

	In sample	Leave one out
Product mix	6.5	4.9
Borrower mix	1.4	0.6
rating	1.5	1.4
size	-0.0	-0.2
age	0.5	0.2
bank debt	-0.5	-0.9
Lender	92.1	94.5

Each share is a component of equation (6) over $\text{var}(\beta_b)$, so the three add to 100. The four borrower rows break $\text{cov}(\beta_b, B_b)$ and add to the borrower line. *Leave one out* re-estimates the slopes once without each lender, and is the column to read.

Neither mix explains much. Product mix accounts for 4.9% of the variance of the gaps, the four borrower margins together 0.6%, and 94.5% is left over. Scoring each lender against slopes its own records helped set raises the two to 6.5 and 1.4%: a lender's own records pull the common slopes toward its own gap, and dropping it takes about

two points of variance back off. The two mixes overlap little. Each alone accounts for 6.9 and 1.3% in sample, which add to 8.3 against the 7.9% they carry jointly. In the units of Figure 1, the remainder has the same standard deviation as the gaps themselves, 0.29 percentage points.

Generating spread and accounting for it are two things. The product prediction has a standard deviation of 0.088 percentage points across lenders, almost a third of the 0.29 the gaps themselves have, and yet 4.9% of their variance moves with it. A mix can push lenders apart in a direction the gaps do not follow, and this one partly does. Re-estimating the lender gaps with the common product gaps entered as controls, rather than subtracting P_b from them, moves the standard deviation from 0.29 to 0.31. That fit adds thirteen parameters to the same records and widens the estimated gaps on its own, so I do not read the rise as dispersion the mix was hiding.

The product mix is small in total, and uneven across products. Its co-movement with the gap splits product by product,

$$\text{cov}(\beta_b, P_b) = \sum_p \gamma_p \text{cov}(\beta_b, w_{bp}), \quad (7)$$

and each product's term, over the total, is the share of the co-movement it carries. The shares add to one by construction. A product's share is positive when the product is dear inside a bundle and concentrated at the dear lenders, or cheap and concentrated at the cheap ones.

Overdrafts carry 65% of that co-movement. A lender's gap correlates 0.31 with its share of cross-sold records in overdrafts, and 0.14 with that share once the common product gaps are held fixed. The margin is coarser than a correlation suggests, because the median lender cross-sells no overdraft at all. The forty-seven lenders above that median average a gap of +0.10 percentage points against -0.05 for the fifty-two at or below it, and net of the mix the two groups sit at +0.17 and +0.11: holding the product gaps fixed cuts the distance between them from 0.15 points to 0.06 but does not close it.

Two more products carry about a quarter each, and one of them through the opposite sign. Revolving credit is dear inside a bundle and concentrated at the dear lenders, which accounts for 22%. Global operating credit is the cheapest well-populated product and the dear lenders cross-sell *less* of it, which moves the covariance the same way and accounts for a further 26%. The three together carry 113% of the co-movement and the rest take 13 points back off, the largest of them equipment loans at -8% . The premium lenders sell more revolving credit, then, and less of the standard operating loan.

Among the borrower margins, rating is the only one above a point, at 1.4%, and bank debt moves against the gap at -0.9% : lenders whose cross-sold borrowers are more leveraged are slightly cheap, not dear, once the other margins are held fixed. Size and age sit within two tenths of a point of zero. The borrower line is smaller than its rating

Table 6: Which products make a lender look dear: the decomposition of equation (7). Across the 99 lenders of Figure 1

	$\bar{w}_p$	γ_p	ρ_p	ρ_p^{net}	Share (%)
Overdraft	0.166	0.224 ^{***}	0.311 ^{***}	0.139	64.7
Global operating credit	0.166	-0.199 ^{***}	-0.123	-0.160	26.3
Revolving credit	0.038	0.221 ^{***}	0.193 [*]	0.111	22.2
Other treasury credit	0.083	-0.063	-0.090	0.016	3.8
Inventory finance	0.014	-0.061	-0.057	-0.016	0.6
Other credit to customers	0.003	-0.399 [*]	-0.022	0.019	0.5
Daily assignment	0.075	0.026	0.007	0.260 ^{***}	0.1
Discounting	0.041	-0.008	0.045	0.253 ^{**}	-0.1
Instalment sales finance	0.006	0.104	-0.034	0.102	-0.8
Unregulated housing	0.032	0.041	-0.089	-0.202 ^{**}	-0.8
Leasing (equipment)	0.011	-0.273 ^{***}	0.016	0.152	-1.4
Foreign receivables	0.017	0.094	-0.123	0.071	-2.9
Personal loans	0.027	0.113 [*]	-0.115	-0.217 ^{**}	-3.7
Other equipment loans	0.317	0.070 ^{***}	-0.132	-0.321 ^{***}	-8.2
All other products	0.004	0.029	-0.058	-0.094	-0.1
Total					100.0

Sorted by the last column. w_{bp} is lender b 's share of its cross-sold records in product p and $\bar{w}_p$ its mean across lenders; γ_p is the common product gap of Table 3. ρ_p correlates w_{bp} with the gap β_b across lenders, ρ_p^{net} the same with the common product gaps held fixed. The last column is the product's term in equation (7) over the total, so it sums to 100. ^{***} $p < 0.01$, ^{**} $p < 0.05$, ^{*} $p < 0.10$, on γ_p and on each correlation.

row because the four margins partly offset.

One borrower margin varies across lenders more than anything inside B_b : whether a lender cross-sells to graded firms at all. The median lender holds 84% of its cross-sold records in ungraded firms, the mean is 71%, and the standard deviation across lenders is 29 points. I regress the lender gap on that share and find nothing. The correlation across the 99 lenders is -0.039 and the slope -0.038 (s.e. 0.100). Moving a lender from a book of entirely graded borrowers to one of entirely ungraded borrowers moves its gap by 0.04 percentage points, against a standard deviation of 0.29 across lenders.

How fast a mix passes into a gap is not measured here. The slope of β_b on the product prediction is 0.78 (s.e. 0.33) in sample and 0.53 (s.e. 0.33) leaving the lender out, and on the rating prediction 0.98 (s.e. 1.65) and -1.07 (s.e. 1.26). On 99 lenders only the in-sample product slope clears zero, and none of the four can be told from one. What the shares settle is the other term: whatever the pass-through rate, the mixes do not vary enough across lenders to move the gaps far.

Together the two mixes account for 5.5% of the variance of the lender gaps. Observable composition is not why these lenders disagree.

4.6 The same firm at several lenders

The comparisons above hold the borrower’s observable characteristics fixed. This one holds the borrower fixed. I absorb a firm $\times$ month effect and take the lender *out* of the cell,

$$\text{TEG}_i = \sum_b \beta_b (\text{CrossSell}_i \times \mathbf{1}[\ell(i) = b]) + \alpha_{f(i) \times t(i)} + \alpha_{c(i)} + \varepsilon_i. \quad (8)$$

The variation is then firms borrowing from two lenders in the same month with a second product at one of them and not at the other. That is rare, and the one-month-in-three sampling makes it rarer. The counts come first, before any coefficient, because a gap estimated off a few hundred firms is a different claim from one estimated off the sample.

Of 911,105 firm-months, 24,587 hold two or more lenders, and 1,791 of those hold both a cross-sold and a standalone record. They are 1,285 firms and 9,089 records.

On that support the average gap disappears. Pooled, $\beta = 0.014$ with a standard error of 0.059 on 8,481 records: a firm taking a second product from one of its banks and not from another pays the same either way.

The lenders still disagree. Twenty-one carry at least 100 identifying records, and their gaps have a standard deviation of 0.56, with $Q = 68$ on 20 degrees of freedom, $p < 0.001$, and $\hat{\tau} = 0.32$ percentage points - larger than the 0.16 of Section 4.1, not smaller. Holding the borrower fixed removes the average and leaves the dispersion, so what separates these lenders is not the firms they lend to.

References

- Adams, W. J. and Yellen, J. L. (1976). Commodity bundling and the burden of monopoly. *Quarterly Journal of Economics*, 90(3):475–498.
- Armstrong, M. and Vickers, J. (2010). Competitive non-linear pricing and bundling. *Review of Economic Studies*, 77(1):30–60.
- Basten, C. and Juelsrud, R. (2025). Monetary policy transmission through cross-selling banks. Working Paper Series 3072, European Central Bank.
- Basten, C. and Juelsrud, R. (2026). Cross-selling in bank-household relationships: Mechanisms and implications for pricing. *Review of Financial Studies*, 39(6):1751–1784.
- Bharath, S., Dahiya, S., Saunders, A., and Srinivasan, A. (2007). So what do i get? the bank’s view of lending relationships. *Journal of Financial Economics*, 85(2):368–419.
- Degryse, H. and Ongena, S. (2005). Distance, lending relationships, and competition. *Journal of Finance*, 60(1):231–266.
- Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *Quarterly Journal of Economics*, 132(4):1819–1876.
- Drechsler, I., Savov, A., and Schnabl, P. (2021). Banking on deposits: Maturity transformation without interest rate risk. *Journal of Finance*, 76(3):1091–1143.
- Drucker, S. and Puri, M. (2005). On the benefits of concurrent lending and underwriting. *Journal of Finance*, 60(6):2763–2799.
- Farrell, J. and Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. In *Handbook of Industrial Organization*, volume 3, chapter 31. Elsevier, Amsterdam.
- Gabaix, X. and Laibson, D. (2006). Shrouded attributes, consumer myopia, and information suppression in competitive markets. *Quarterly Journal of Economics*, 121(2):505–540.
- Ioannidou, V. and Ongena, S. (2010). Time for a change: Loan conditions and bank behavior when firms switch banks. *Journal of Finance*, 65(5):1847–1877.
- Kanatas, G. and Qi, J. (2003). Integration of lending and underwriting: Implications of scope economy. *Journal of Finance*, 58(3):1167–1191.
- Klemperer, P. (1995). Competition when consumers have switching costs: An overview with applications to industrial organization, macroeconomics, and international trade. *Review of Economic Studies*, 62(4):515–539.

- Lepetit, L., Nys, E., Rous, P., and Tarazi, A. (2008). The expansion of services in european banking: Implications for loan pricing and interest margins. *Journal of Banking and Finance*, 32(11):2325–2335.
- McAfee, R. P., McMillan, J., and Whinston, M. D. (1989). Multiproduct monopoly, commodity bundling, and correlation of values. *Quarterly Journal of Economics*, 104(2):371–383.
- Nalebuff, B. (2004). Bundling as an entry barrier. *Quarterly Journal of Economics*, 119(1):159–187.
- Ongena, S., Paraschiv, F., and Reite, E. J. (2021). Determinants of price discrimination and switching mortgage provider in times of regulation and digitalization. Working paper, University of Zurich.
- Whinston, M. D. (1990). Tying, foreclosure, and exclusion. *American Economic Review*, 80(4):837–859.

A Appendix: the lenders outside the cross-section

Section 4.1 estimates one cross-selling gap per lender and reports the 99 lenders that clear a record floor. The same estimates support two other lender sets, and this appendix gives both: every lender the sample can estimate, and the twenty largest by volume. All three come from one run, so they differ in which lenders they keep and in nothing else - same sample, same comparison cell, same clustering.

A.1 Every lender the sample can estimate

Dropping the floor adds 18 lenders, for 117 - every lender whose cross-selling gap the sample can estimate at all. Figure A.1 draws them.



Figure A.1: The cross-selling gap at every lender the sample can estimate

Notes. One point per lender, from the lender-by-lender form of equation (1); spikes are 95% confidence intervals on standard errors clustered by firm. The floor described in the text is not imposed here. The 99 lenders that clear it are Figure 1.

The floor removes noise, not disagreement. The 18 added lenders widen the observed spread by more than half, from a standard deviation of 0.29 to 0.45, because each is estimated on fewer records. They leave the dispersion of the underlying gaps where it was: $\hat{\tau} = 0.16$ on the 117, the same 0.16 as on the 99, and the weighted mean stays at -0.05 . Cochran's Q is 631 on 116 degrees of freedom, $p < 0.001$, and $I^2 = 82\%$. Adding the borrower as a fixed effect gives $\hat{\tau} = 0.086$ on the 108 lenders whose standard error survives, against 0.083 on 93 - in each arm the remainder are excluded because the firm-clustered sandwich collapses to numerical zero and Q weights by $1/se^2$.

What the floor removes is seventeen lenders indistinguishable from zero and one carrying a premium. Both arms count the same fifteen significant discounts, so the discount

tail of Figure 1 is the whole discount tail of the market. The added lenders are the thinner ones: the median gap here rests on 646 contributing firms against 778 in the body’s arm. Across the 117, the comparison cell leaves a median 81% of a lender’s records contributing, and 7 lenders lose more than half.

A.2 The twenty largest lenders

The twenty largest lenders by volume are the set the earlier design used (Appendix D), so this arm says what the body’s wider cross-section adds. Figure A.2 draws them on the pooled sample and the cross-selling definition, so the only thing that changes against Figure 1 is which lenders are kept.

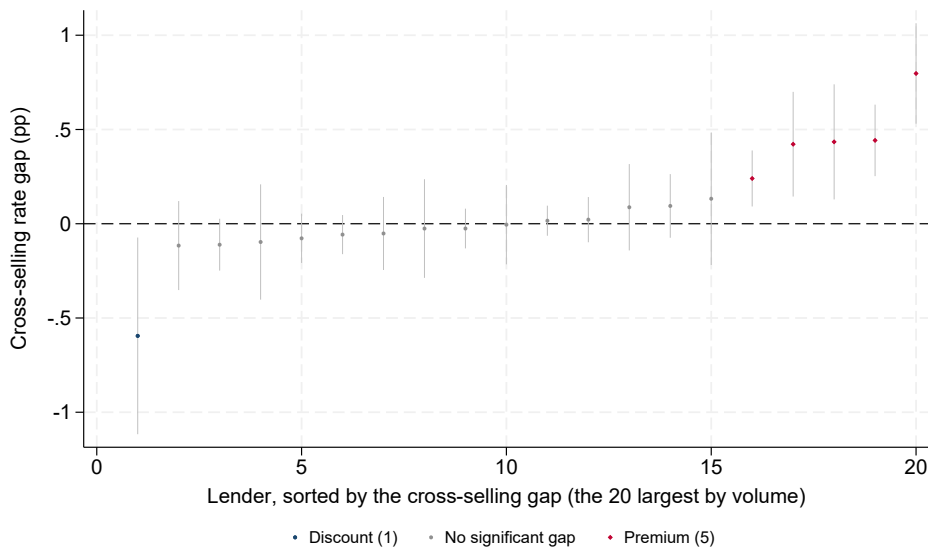


Figure A.2: The cross-selling gap at the twenty largest lenders

Notes. As Figure A.1, restricted to the twenty largest lenders by number of records. All twenty clear the floor of Section 4.1.

The largest lenders disagree too, and they charge a premium. Their gaps have a standard deviation of 0.26 against 0.29 across the 99, $Q = 84$ on 19 degrees of freedom with $p < 0.001$, $I^2 = 77\%$ and $\hat{\tau} = 0.15$ against 0.16. So the disagreement in Section 4.1 is not something small lenders generate. What does change is the level: the weighted mean gap is $+0.034$ here against -0.05 across the 99, and none of the twenty shows a discount against 15 of 99. Cross-selling is dearer at the large lenders and cheaper at the rest.

The within-borrower disagreement is not in this set. Holding the borrower fixed inside each lender gives $\hat{\tau} = 0.008$ with $p = 0.43$: across the twenty largest lenders, the gap estimated on the same firm is consistent with one common number. On the 99 the same quantity is 0.083 with $p < 0.001$. Precision is not the explanation: a median 1,290 firms stand behind each of the twenty gaps against 778 in the body’s arm, so these

estimates are the better measured ones and the dispersion is smaller. The within-borrower result of Section 4.1 therefore comes from the lenders outside the top twenty, and the wider cross-section is what makes it available.

B Appendix: which borrowers pay

The gaps of Section 4.1 are lender averages. This section asks whether, inside a lender, the gap falls on some borrowers more than on others. Interact the treatment with the borrower’s characteristics inside the same cell,

$$\begin{aligned} \text{TEG}_i = & \beta \text{CrossSell}_i + \sum_{g \neq g_0} \gamma_g (\text{CrossSell}_i \times \mathbf{1}[g(i) = g]) \\ & + \sum_{g \neq g_0} \mu_g \mathbf{1}[g(i) = g] + \alpha_{\ell(i), c(i)} + \varepsilon_i, \end{aligned} \tag{9}$$

so γ_g is the extra gap group g pays against the base group at the same lender in the same month. Standard errors cluster on lender and firm. Asking it in one step on the record data, rather than pricing each bundle and regressing on that price, keeps the inference honest: a fitted bundle price is a generated regressor and its second-stage standard errors understate. Bank debt over total assets is winsorised at 1 and 99 and cut into quartiles, with a fifth bin for firms filing no balance sheet; credit quality carries a fifth level for firms the agency does not grade, and age a sixth bin for firms with no recorded creation date.

Table B.1: Who pays for cross-selling, equation (9).

Dependent variable: TEG, in percentage points

	γ_g	s.e.	Cross-sold
<i>Bank debt over total assets</i>			
Q1 lowest	base		2,969
Q2	-0.215	0.139	5,613
Q3	-0.061	0.158	6,392
Q4 highest	-0.122	0.192	4,666
No balance sheet	-0.196	0.169	51,041
<i>Credit quality</i>			
Excellent	-0.131	0.123	1,712
Good	base		7,357
Medium	-0.095*	0.053	9,288
Weak	0.166	0.145	1,936
Not rated	-0.086	0.073	50,388
<i>Firm age</i>			
0-2y	-0.112	0.069	7,828
3-5y	0.004	0.056	6,166
6-10y	-0.001	0.042	10,207
11-20y	base		16,611
>20y	-0.011	0.032	25,416
No creation date	-0.291***	0.080	4,453

Each γ_g is against the base group of its own margin. Groups with fewer than 100 cross-sold records are not reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

What a borrower is does not move the gap; whether it is documented does. Bank dependence gives -0.215 , -0.061 and -0.122 across the upper three quartiles, none of them distinguishable from zero, none of them monotone, and a joint p of 0.13. Credit quality gives a joint p of 0.44: the weakest graded firms pay 0.166 more (s.e. 0.145), the excellent 0.131 less (s.e. 0.123), the ungraded 0.086 less (s.e. 0.073), and only the medium class clears the ten percent level, at -0.095 (s.e. 0.053). Age gives a joint p of 0.0003, and it comes from a single bin: firms with no recorded creation date pay 0.291 less (s.e. 0.080), while the five genuine age bands differ from the base group by at most 0.11 and none of them clears zero. Firms filing no balance sheet sit the same way on the bank-dependence margin, at -0.196 , though the standard error of 0.169 does not separate them from the base group.

Within a lender, cross-selling costs the same whoever the borrower is. What separates the lenders of Section 4.1 is not the kind of firm each one lends to.

C Appendix: controlling for every observable in the register

The matching cell of equation (1) holds the lender, the month, the product, the amount band, the maturity band and the credit class fixed. Everything else the register records about a borrower or a contract stays in the error term. I ask here how much of the cross-lender dispersion survives once I put all of it on the right-hand side.

I re-estimate the equation lender by lender under four further rungs, each cumulative on the last. The first adds the terms of the contract: log amount, log maturity, the borrower’s share of the lending pool, and the shares of the record at a variable rate, on a revolving line and under a state scheme. The second adds the balance sheet: assets, turnover, employment, bank debt, financial debt, short-term risk, the confirmed and documentary margins, guarantees, credit drawn and credit available, the drawn share, bank-debt dependence, utilisation and firm age. The third adds the categorical record: the fine rating, the size class, the age band, the bank-dependence quartile, region, department, group membership, foreign location, residence zone, the size code, the beneficiary code and the sole-proprietorship flag. The fourth adds the four-digit sector and the lending branch. A register column enters unless it is an alternative dependent variable, an identifier, or constant over the sample period.

Each control carries its own bin for a missing value, so no record leaves the sample for want of a column, and the first four rungs run on the same records. The fourth loses a few, because its finest cells hold no comparison. The last row of the table therefore re-estimates the baseline on the records the fourth rung keeps, which separates what the controls do from what the lost records do.

Table C.1: The lender cross-section under progressively richer controls, the gaps β_b of equation (1).

Dependent variable: TEG, the all-in rate, in percentage points

	Pooled gap	SD	Corr.	In Q	$\hat{\tau}$	Q
Matching cell of equation (1)	0.065 (0.013)	0.29	1.00	99	0.16	610
+ contract terms	0.024 (0.012)	0.23	0.85	98	0.13	481
+ borrower balance sheet	0.049 (0.012)	0.23	0.80	96	0.10	337
+ borrower categories and location	0.041 (0.012)	0.22	0.77	95	0.092	311
+ fine sector and lender branch	0.026 (0.012)	0.26	0.73	92	0.090	292
Matching cell, on the records of the row above	0.065 (0.013)	0.29	0.99	99	0.16	608

Each row adds the controls it names to the matching cell of equation (1). The lender columns describe the 99 gaps of Figure 1; the pooled column is one regression on all 208 lenders with the lender inside the cell. In Q counts the gaps the test can weight. Standard errors are clustered at the firm level.

The controls take a bit under half of the dispersion, and the lenders still do not share one gap. $\hat{\tau}$ falls from 0.16 percentage points to 0.090. The largest step is the contract terms, which take it to 0.13, and the next is the balance sheet, which takes it to 0.10; the categorical record then takes it to 0.092 and the sector to 0.090. Cochran's Q falls with it, 610, 481, 337, 311 and 292 on 98, 97, 95, 94 and 91 degrees of freedom, with $p < 0.001$ at every rung, and I^2 falls from 84% to 69%. Twenty lenders have a gap that clears zero at the 95% level at the richest rung, eleven at a discount and nine at a premium, against thirty-four at the baseline.

The degrees of freedom fall along the ladder because the test loses lenders, not because the ladder stops estimating them: every rung returns all 99 gaps. One lender at the first rung, three at the second, four at the third and seven at the fourth come back without a standard error. Inside those lenders the controls and the absorbed cell together leave the cluster-robust variance matrix singular, and the fit returns no standard error for any coefficient in it. Q weights by the inverse squared standard error, so it drops them.

The raw standard deviation does not fall in step: 0.29, 0.23, 0.23, 0.22, and then back up to 0.26. Finer cells leave fewer records on each side of the comparison and noisier lender estimates, which widens the raw spread while $\hat{\tau}$, which nets estimation error out, keeps falling. The same noise shows in the correlation with the baseline gaps, which falls a step at a time: 0.85, 0.80, 0.77, 0.73.

The pooled column asks the other question: not whether the lenders still disagree, but whether the cross-selling indicator still has variation of its own once the controls are in. I run one regression on all the records at once with the lender inside the absorbed cell, so the gap compares a cross-sold record with a single-product record at the same lender in the same month, product, amount band, maturity band and credit class, and reports one number for the market instead of 99. That column uses every lender rather than the 99 the floor keeps, and 946,759 of the 1,087,737 records of Section 3; the rest sit in cells with nothing to compare them to. The gap is 0.065 (0.013) at the matching cell, 0.024 (0.012) with the contract terms, 0.049 (0.012) with the balance sheet, 0.041 (0.012) with the categorical record and 0.026 (0.012) with the sector and the branch. The standard error stays at 0.012 throughout, and the smallest gap on the ladder, 0.024 with the contract terms, is twice it. The controls do not absorb the cross-selling indicator. They move the lenders towards each other.

The last row rules out the sample. Re-estimating the baseline on the records the richest rung keeps gives a standard deviation of 0.29 again, $\hat{\tau} = 0.16$, $Q = 608$ on 98 degrees of freedom, a correlation of 0.99 with the baseline itself and a pooled gap of 0.065 (0.013), on a median of 4,011 records per lender against 4,118. What the ladder removes is the work of the controls, not of the records.

The ladder ends where the firm fixed effects of Section 4.1 end: $\hat{\tau}$ is 0.090 here and 0.083 there. Every observable the register carries about a borrower and the borrower's

own identity remove the same share of the dispersion, and neither removes the rest.

Two readings fit this. On the first, part of what [Section 4.1](#) assigns to the lender is borrower and contract composition the matching cell does not hold fixed, and the dispersion to quote is 0.090 rather than 0.16. On the second, the controls doing most of the work are themselves priced: the amount, the maturity and the share of the pool are agreed with the rate in one negotiation, so conditioning on them removes variation the lender creates rather than variation it faces. Equation (1) keeps them out of the conditioning set for that reason, and bins amount and maturity instead. Which reading is right changes the size of the dispersion and not its existence: with every column the register carries in the regression, the ninety-nine lenders do not share one gap.

D Appendix: the earlier design on the complete register

This appendix reports an earlier version of the exercise, kept because it carries two things the design in the body cannot. It defines a bundle as *any two contracts* from the same bank in the same month rather than two distinct products, and it runs on the complete register, 2015q1–2022q3, with no restriction on period, subsidised status or product. The quantity it estimates is therefore not the cross-selling gap of Section 4, and its numbers are not comparable to that section’s: roughly three fifths of the multiplicity it treats is one product drawn repeatedly, which is the reason the body does not use this definition. What it adds is the complete register, and a lender cross-section of 129 built under its own sample rule.

D.1 The complete sample

The complete register runs 2015q1–2022q3 and contains 4,851,195 contracts, with no restriction on period, subsidised status or product. Table D.1 reports the pooled bundle gap under three absorbed cells.

Table D.1: Pooled bundle gap on the complete sample, three absorbed cells

	Absorbed cell	Gap (pp)	s.e.	N
A	bank $\times$ month $\times$ product $\times$ amount $\times$ maturity	−0.241 ^{***}	0.083	4,765,749
B	bank $\times$ month $\times$ product $\times$ rating	−0.193 ^{***}	0.067	4,807,456
C	A + rating	−0.105 [*]	0.055	4,521,786

SEs two-way clustered by bank and firm. Unrated firms enter as an explicit rating category, so absorbing the rating drops nothing. Marks as in the body tables: ^{***} $p < 0.01$, ^{**} $p < 0.05$, ^{*} $p < 0.10$.

On the complete sample the pooled gap is −0.105pp with $t = 1.9$ - marginal. Adding the credit rating to a cell that already matches on contract terms removes 56% of the raw gap, the first indication that borrower composition rather than pricing drives much of the apparent discount. An independent route agrees: absorbing firm fixed effects instead of the rating gives −0.125pp.

D.2 All 129 lenders

Estimating structure C separately for every lender with at least 5,000 rated contracts and at least 200 in each bundle state yields 129 lenders. The median lender’s gap is −0.026pp, against a 10th percentile of −0.534 and a 90th of +0.359. **Forty-one lenders show a significant discount and twenty-two a significant premium**; the remaining sixty-six are indistinguishable from zero.

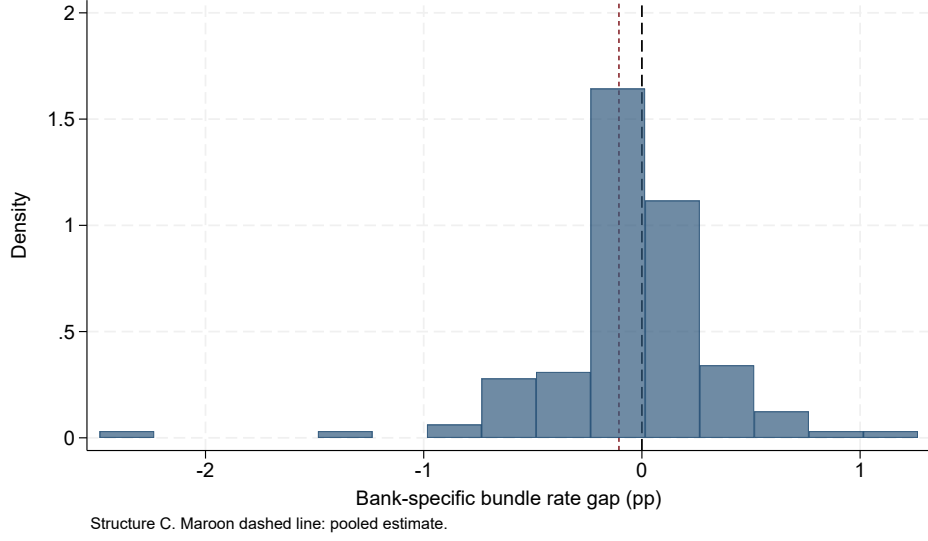


Figure D.1: Distribution of the bundle gap (structure C) across the 129 lenders meeting the sample rule, complete sample.

Table D.2: The twenty largest lenders on the complete sample

Rank	Bank	Volume (%)	Bundle share (%)	Gap (pp)	s.e.	N
1	Bank A	10.63	52.2	0.083	0.057	10,407
2	Bank B	6.04	14.3	-0.580	0.039	28,186
3	Bank C	5.41	16.3	-0.059	0.016	104,664
4	Bank D	4.52	30.0	-0.578	0.017	92,464
5	Bank E	4.22	98.7	0.148	0.025	51,038
6	Bank F	4.08	13.7	0.084	0.086	22,518
7	Bank G	4.00	23.4	-0.152	0.081	63,076
8	Bank H	3.11	65.1	-0.088	0.073	6,049
9	Bank I	2.19	91.0	-0.099	0.014	667,382
10	Bank J	2.14	80.6	0.424	0.076	32,885
11	Bank K	2.08	23.7	-0.280	0.094	13,968
12	Bank L	2.02	14.4	-0.272	0.113	12,427
13	Bank M	1.66	23.1	0.063	0.076	29,686
14	Bank N	1.54	77.3	-0.085	0.036	30,784
15	Bank O	1.51	57.4	0.902	0.055	17,525
16	Bank P	1.50	94.2	-0.783	0.297	282,920
17	Bank Q	1.23	13.8	-0.204	0.082	13,802
18	Bank R	1.13	87.7	-0.447	0.036	364,038
19	Bank S	0.99	11.0	0.158	0.112	24,665
20	Bank T	0.97	15.3	-0.037	0.072	11,657

Complete sample, structure C. Lenders are anonymised by volume rank; the labels are the same institutions as in Table D.5, though the rank order differs because that table ranks within the restricted window.

D.3 Which controls matter, and which do not

Table D.3 re-estimates each of the twenty largest lenders’ gaps under a ladder of progressively finer comparison cells, on the complete sample. Rung S3 reproduces the published structure-C estimate exactly, so the remaining rungs are interpretable relative to it.

Table D.3: Cross-lender dispersion under progressively finer cells

Rung	Absorbed cell	SD	p10	p50	p90	Median cells
S1	month $\times$ product	0.617	−1.176	−0.130	0.225	171
S2	+ amount $\times$ maturity (<i>published A</i>)	0.445	−0.728	−0.127	0.208	772
S3	+ rating (<i>published C</i>)	0.372	−0.579	−0.086	0.291	1,058
S4	+ sector	0.363	−0.486	−0.063	0.438	924
S5	+ firm size	0.405	−0.473	−0.064	0.589	626
S6	+ firm fixed effects	0.178	−0.063	0.029	0.263	1,058

Complete sample, twenty largest lenders. “Median cells” is the median across lenders of the number of comparison cells containing both a bundled and a solo contract; cells containing only one state contribute nothing to a within estimator. S6 is counted on the S3 partition, since it asks what firm fixed effects add on top of S3.

Observable borrower characteristics do almost nothing; the firm identity does everything. Adding sector (S4) moves the cross-lender dispersion from 0.372 to 0.363, and adding a decile of total assets (S5) moves it to 0.405 - it rises, because the finer cells leave fewer contributing observations and noisier estimates, with the median lender losing roughly 40% of its comparison cells. Neither changes a single lender’s sign, and both correlate above 0.95 with S3. Only firm fixed effects (S6) move the picture: the dispersion halves to 0.178, the correlation with S3 falls to 0.305, and six of twenty lenders change sign.

The natural reading of S5’s null is not that firm size is irrelevant but that the control is too coarse to capture it: assets are missing for a large share of small firms and enter as a single residual bin, whereas firm fixed effects absorb size however it is measured. The imbalance is real - bundled borrowers are roughly three and a half times larger than solo borrowers within the same lender - so a control that fails to remove it is better read as weak than as evidence of no effect.

D.4 Bundle composition across all 129 lenders

Table D.4 is the full-cross-section counterpart of Table D.6, and carries the statistical weight that the restricted version cannot: it compares 41 discount lenders against 22 premium lenders rather than 8 against 3.

Table D.4: Bundle composition by pricing type, all 129 lenders

Share of bundles containing	Discount (41)	Premium (22)	p
Overdraft	0.196	0.468	0.000
Treasury	0.297	0.080	0.002
Leasing	0.246	0.024	0.010
Equipment	0.317	0.216	0.185
Consumer / revolving	0.116	0.203	0.112
Trade / export	0.150	0.162	0.865
Property	0.042	0.083	0.319
Other	0.006	0.002	0.198
Subordinated	0.000	0.000	0.590
Repeat (same product)	0.633	0.711	0.311
Cross-sell (2+ products)	0.367	0.289	0.311

Complete sample. Share of each lender’s bundled contracts whose firm–lender–month contains the stated family, averaged within pricing type; p -values from two-sample t -tests across lenders.

A bundle may contain several families, so shares do not sum to one.

Overdraft content is the sharpest separation in the table and the strongest of any lender characteristic tested anywhere in this paper: premium lenders’ bundles contain an overdraft 46.8% of the time against 19.6% for discount lenders ($p < 0.001$). Treasury credit and leasing run the other way. **The repeat versus cross-sell distinction does not separate the groups at all** ($p = 0.311$), which is worth stating explicitly because it is the natural first hypothesis: the difference between the two groups is which *products* they bundle, not whether they are cross-selling or drawing one product repeatedly.

D.5 Lender-level detail on the restricted sample

Table D.5 gives the lender-by-lender estimates under this design’s own restricted window, 2016–2019 and the twenty largest lenders, under structure C and with firm fixed effects added. Section 4.1’s per-lender estimates are a different object on a different sample, and they are in Figure 1. Table D.6 reports the bundle composition of premium and discount lenders on the restricted sample; it is the underpowered counterpart of Table D.4, with three premium lenders in the strict sample and two in the wide, and is reported for completeness rather than inference.

Table D.5: The bundle gap, lender by lender, restricted sample

Rank	Bank	Structure C			+ firm FE			Type
		Gap	s.e.	N	Gap	s.e.	N	
1	Bank A	-0.027	0.062	5,326	-0.052	0.021	4,814	–
2	Bank B	-0.824	0.052	12,965	-0.315	0.066	5,129	Discount
3	Bank C	-0.115	0.032	38,316	-0.023	0.063	17,434	Discount
4	Bank F	0.156	0.058	9,267	-0.072	0.065	3,025	Premium
5	Bank D	-0.640	0.022	37,184	-0.018	0.011	29,606	Discount
6	Bank G	0.038	0.045	9,646	0.023	0.066	5,124	–
7	Bank H	-0.298	0.102	3,316	0.019	0.048	3,185	Discount
8	Bank K	-0.377	0.115	7,439	0.095	0.076	5,157	Discount
9	Bank L	-0.040	0.077	2,249	0.002	0.059	1,270	–
10	Bank N	-0.118	0.037	21,197	0.006	0.016	20,181	Discount
11	Bank P	-0.542	0.117	148,768	0.232	0.218	141,266	Discount
12	Bank M	0.073	0.121	16,521	0.168	0.103	11,723	–
13	Bank O	0.874	0.060	10,739	0.251	0.084	7,299	Premium
14	Bank J	-0.149	0.047	18,221	-0.033	0.017	17,701	Discount
15	Bank Q	-0.023	0.040	2,306	0.049	0.070	1,002	–
16	Bank S	0.190	0.146	14,029	0.156	0.110	9,242	–
17	Bank U	-0.081	0.069	12,872	-0.163	0.109	8,139	–
18	Bank V	-0.034	0.084	22,042	0.087	0.086	15,430	–
19	Bank W	0.215	0.069	2,951	0.056	0.046	1,067	Premium
20	Bank T	-0.083	0.115	6,011	-0.007	0.192	3,830	–

Strict sample. Lenders are anonymised and ordered by volume rank within the restricted window; the same labels identify the same institutions in Table D.2. Type is the classification from the structure-C interval.

Table D.6: Bundle composition by pricing type, restricted sample

Share of bundles containing	Strict			Wide		
	Disc.	Prem.	diff	Disc.	Prem.	diff
Overdraft	0.157	0.298	+0.141	0.213	0.448	+0.234
Equipment	0.138	0.343	+0.204	0.233	0.342	+0.108
Treasury	0.349	0.360	+0.012	0.357	0.247	−0.110
Trade / export	0.332	0.199	−0.133	0.295	0.021	−0.274
Leasing	0.074	0.000	−0.074	0.049	0.000	−0.049
Consumer / revolving	0.039	0.010	−0.029	0.076	0.012	−0.064
Property	0.033	0.025	−0.008	0.028	0.034	+0.006
Repeat (same product)	0.851	0.738	−0.114	0.727	0.894	+0.167
Cross-sell (2+ products)	0.149	0.262	+0.114	0.273	0.106	−0.167

Share of each lender's bundled contracts whose firm–lender–month contains the stated family, averaged within pricing type. A bundle may contain several families, so columns do not sum to one. No significance test is reported: with two and three lenders none would be meaningful.

D.6 The firms behind the within-firm estimates

Equation (8) is identified only from firms borrowing from two or more lenders in the same month with both bundle states present. Panel A counts them at each level of identification; Panel B compares them with the firms that never use more than one lender. The counts are this design's, on the complete register and the any-two definition; Section 4.6 gives the counts that identify the equation as the body estimates it.

Table D.7: Who identifies the within-firm estimates*Panel A. How many firms identify each specification*

Level of identification	Firms	Share	Contracts
All firms in the estimation sample	127,866	1.000	425,889
2+ contracts (identifies a firm FE)	47,210	0.369	345,233
2+ lenders ever	6,502	0.051	106,253
2+ lenders in the same month	2,432	0.019	83,633
...and both bundle states	865	0.007	69,202

Panel B. Who they are

Characteristic	One lender	2+ lenders	Identifying	<i>p</i>
Firm age (years)	14.92	25.65	30.80	0.000
Mean size band (1=A .. 13=N)	7.65	5.84	4.72	0.000
Share with a size band	0.291	0.772	0.874	0.000
Mean log assets	7.97	9.44	10.78	0.000
Mean log turnover	8.18	9.20	9.87	0.000
Mean employees	73.1	205.0	250.4	0.000
Share good-rated	0.477	0.558	0.613	0.000
Share weak-rated	0.523	0.442	0.387	0.000
Share unrated	0.728	0.251	0.150	0.000
Mean cotation (1 best, 12 worst)	5.49	5.05	4.83	0.000
Contracts in the window	2.63	16.34	80.00	0.000
Share of contracts bundled	0.089	0.223	0.614	0.000

Restricted sample, top twenty lenders, so “two lenders” means two of those twenty. One row per firm, so a firm with many contracts does not dominate. *p*-values from two-sample *t*-tests comparing firms with two or more lenders against firms with one.

The identifying firms are a small and highly selected group: 865 of 127,866, or 0.7%. Because they borrow far more often than the rest - 80 contracts each against 2.6 - they nonetheless account for 69,202 contracts, 16% of the sample.

They differ from the rest on every characteristic measured, all at $p < 0.001$. They are twice as old (31 against 15 years), roughly sixteen times larger in assets, carry a BdF size band 87% of the time against 29%, are unrated only 15% of the time against 73%, and bundle 61% of their contracts against 9%. In short they are the large, long-established, well-documented corporates that maintain several banking relationships at once.

The bound on external validity runs in an awkward direction: the firms for which the sharpest test is available are precisely those about which most is known, while Appendix B finds the gap moves with whether a borrower is documented at all. Whether the same conclusion holds for opaque single-banked borrowers cannot be established from this

register.

D.7 What the restrictions change, and what they do not

Table D.8: Complete versus restricted sample, side by side

	Complete 2015–2022, all lending	Restricted (strict) 2016–2019, top 20
Contracts	4,851,195	1,576,148
Lenders estimated	129	20
Pooled gap, structure C	−0.105	-
SD across lenders, structure C	0.372 [†]	0.355
SD across lenders, + firm FE	0.178 [†]	0.129
Ratio	0.48	0.36
I^2	98.7% [†]	98.2%

[†] computed on the twenty largest lenders, so that the two columns are comparable.

The restrictions remove two thirds of the register and leave the central facts intact: the dispersion across lenders is far larger than sampling error in both samples, firm fixed effects roughly halve it in both, and the reduction falls almost entirely on the discount tail in both. Both columns are this appendix’s design, so the comparison is between its two samples; the body’s sample is narrower than either, because its bundle gaps are cleanly interpretable as lender pricing - no administered rates, no state-guaranteed product, no Covid quarter.